One-Day-Ahead Value-at-Risk Forecasting for SPY

A Frozen Out-of-Sample Study of Historical Simulation, GARCH-t, HAR Quantile Regression, and Small Neural Quantile Models

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1 Executive Summary

This report studies one-day-ahead Value-at-Risk (VaR) forecasting for SPY log returns at the 1%, 5% and 10% lower tails, under a strictly causal rolling-window protocol. Six model families are compared on a frozen out-of-sample period starting 2008-01-01: rolling Historical Simulation (HS), GARCH(1,1) with Student-t innovations, GJR-GARCH(1,1,1)-t, linear (HAR-style) quantile regression, a small multi-quantile MLP, and a small GRU quantile model. Feature ablations isolate the incremental value of realized volatility (rv5), bipower variation (bv), and jump/downside-asymmetry information. All numbers in this report are regenerated by scripts from machine-readable prediction artifacts; none are hand-entered.

2 Problem Definition

For each forecast origin t we predict the conditional return quantile $q_{\alpha,t+1}$ with $P(r_{t+1} \leq q_{\alpha,t+1} \mid \mathcal{F}_t) = \alpha$ for $\alpha \in \{0.01, 0.05, 0.10\}$, using only information available at the end of day t. A violation is defined as $r_{t+1} \leq \hat{q}_{\alpha,t+1}$ (VaR is the return quantile itself; lower-tail values are typically negative).

3 Data

Daily SPY observations with columns `log_ret`, `rv5` and `bv`. `rv5` is the realized-volatility measure provided with the assignment; its numerical scale in this sample is variance-like (so that $\sqrt{\texttt{rv5}}$ is close to the daily volatility scale), which motivates considering both $\log(\texttt{rv5})$ and $\sqrt{\texttt{rv5}}$ as structured predictors. `bv` is the bipower-variation counterpart of the same realized-measure family. See Appendix B for the machine-generated data audit.

4 Exploratory Analysis

See Figures 1–2. Key features: volatility clustering, heavy tails (excess kurtosis), strong persistence of log realized measures, and a positive link between negative returns and next-day volatility.

5 Experimental Protocol

Development period: all dates before 2008-01-01 (rolling-origin validation over 2005–2007, with a documented slight adjustment for the 1500-day window whose earliest feasible origin is late 2005).

Final test: all dates from 2008-01-01. Candidate windows: 1000 and 1500 trading days only; the primary window is chosen from development evidence (see Section 11). All models share the same forecast dates. The final configuration and data hash are frozen in `docs/FREEZE_MANIFEST.md` before any final-test run; the gate is enforced by tests and scripts.

6 Computational Setup

All experiments were run under Python 3.11.15 with 32 process workers and BLAS/OpenMP threads limited to one per worker. All reported final models were executed on cpu; the neural models are deliberately small. The 12 primary frozen panels required 1699 seconds in aggregate, with the most expensive individual specification being M5 (about 812 s). Package versions and per-experiment runtimes are recorded in the prediction manifests (../outputs/runs/e34928235ed6-058d5aea/predictions/*.manifest.json).

7 Models

7.1 M0 – Historical Simulation

The empirical quantile of the rolling-window returns: $\hat{q}_\alpha = \hat{F}_t^{-1}(\alpha)$ over $\{r_{t-W+1}, \dots, r_t\}$. Non-parametric benchmark; implicitly assumes a stationary return distribution within the window, so regime shifts are absorbed only with a lag.

7.2 M1 – GARCH(1,1)-Student-t

Classical conditional volatility model with conditional variance

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \qquad \varepsilon_t = \sigma_t z_t, \tag{1}$$

where z_t is an i.i.d. unit-variance standardized Student-t innovation with ν degrees of freedom (variance 1, so that $\mathrm{Var}(\varepsilon_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$). The one-day-ahead VaR is

$$\mathrm{VaR}_{\alpha,t+1} = \mu + F_t^{-1}(\alpha;\nu)\,\hat{\sigma}_{t+1}, \tag{2}$$

with $F_t^{-1}(\alpha;\nu)$ the quantile of the standardized Student-t (implemented via `arch.univariate.StudentsT`, equal to $t_\nu^{-1}(\alpha)\sqrt{(\nu-2)/\nu}$); μ is a constant mean. Parameters are re-estimated by MLE on every rolling window.

7.3 M4 – GJR-GARCH(1,1,1)-Student-t

Adds a leverage term to the variance recursion,

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \gamma\,I(\varepsilon_{t-1} < 0)\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \tag{3}$$

capturing asymmetric responses of volatility to negative shocks; VaR construction identical to M1. Serves as the downside-asymmetry robustness extension.

7.4 M2 – Linear / HAR Quantile Regression

Direct conditional quantile estimation by minimizing the pinball loss,

$$\hat{\beta}_\tau = \arg\min_\beta \sum_t \rho_\tau\big(y_{t+1} - x_t^\top \beta\big), \qquad \rho_\tau(u) = u\big(\tau - I(u < 0)\big), \tag{4}$$

fit separately for $\tau \in \{0.01, 0.05, 0.10\}$ on the same feature sets as the MLP (so the linear-vs-nonlinear contrast is not confounded by information). Independent fits may cross; the crossing rate is reported.

7.5 M3 – Multi-Quantile MLP

A small MLP with joint pinball loss over the three tails and a structurally non-crossing head:

$$q_{0.01} = z_1, \qquad q_{0.05} = z_1 + \text{softplus}(z_2), \qquad q_{0.10} = q_{0.05} + \text{softplus}(z_3), \tag{5}$$

so that $q_{0.01} \le q_{0.05} \le q_{0.10}$ holds by construction (softplus > 0). Targets are standardized with train-only statistics inside each rolling window and the affine transform is inverted on output (Scheme B, audit fix). Training uses early stopping on the last 10% of the window in time order; scalers are fitted on training rows only.

7.6 M5 – GRU Quantile

A one-layer GRU over the last 22 daily feature vectors with the same ordered head and loss. The role is a robustness extension: it tests whether an explicit sequence representation adds value beyond the structured HAR-style features already present, not a substitute for the MLP. Parameters are deliberately kept small.

7.7 Gaussian-GARCH diagnostic

GARCH(1,1) with Gaussian innovations is estimated only in development as a diagnostic reference.

8 Feature Engineering

Information sets (identical for M2 and M3, so Linear-vs-MLP differences are attributable to the mapping rather than the inputs): **F0** returns only (lags 1/2/5/22, abs, squared); **F1** adds $\log rv5$, $\sqrt{rv5}$, 5-day and 22-day HAR aggregations of $\log rv5$; **F2** adds the same block for bv; **F3** adds jump proxy $\max(rv5 - bv, 0)$, relative jump, and a 5-day downside-shock intensity. All features are causal by construction and verified by truncation-invariance tests.

9 Rolling Forecast Design

A unified rolling engine controls window boundaries, feature cutoffs, preprocessing (scalers fitted on the window training rows only), model refits (daily full refit for every model), and prediction recording. Early stopping for neural models uses the last 10% of the window in time order. All outputs share one schema (RESULTS_SCHEMA.md) with `fit_status` logging; failures are never silently dropped.

10 VaR Backtesting Methodology

Per model and tail: empirical failure rate, Kupiec unconditional coverage test, Christoffersen independence and conditional-coverage tests, mean pinball loss, quantile-crossing rate, violation clustering, Dynamic Quantile test (reported as an additional diagnostic), Diebold–Mariano tests and moving-block bootstrap for pairwise loss comparisons. Finite-sample caveats apply: the 1% tail has very few expected violations even in a decade-long test, so coverage tests have low power and small numerical differences must not be over-read.

11 Development / Validation Results

Window comparison tables: `outputs/development/window_selection.csv` and `window_selection_cells.csv`. Decision rule (predeclared, audit-fixed aggregation): per (model, feature-set, tail) cell the candidate-relative normalized loss (cell loss divided by the cell mean across the two candidates) is averaged with equal weight across all cells; per-cell winners, pairwise win counts and drop-one sensitivity are reported alongside. If the equal-weight aggregate and the raw pinball-sum aggregate disagree, the longer window is chosen conservatively for 1% tail stability. The longer window provides more effective tail observations for the 1% quantile (about 15 vs 10 expected violations) and hence more stable empirical quantiles, at the cost of slower regime adaptation; the short window adapts faster. The primary window selected from development evidence is `1500`.

12 Frozen Out-of-Sample Results

All models evaluated on `2641` identical forecast dates ($\geq$ 2008-01-01). Primary seed `42`. Table 1 reports coverage and loss statistics.

13 Tail-Level Model Comparison

See Figure 7. Rankings are tail-specific; a single universal winner is not assumed.

14 Feature Ablations

Table 2 and Figure 8. Ablation interpretation is confined to M2/M3 across F0–F3.

15 Crisis and Regime Analysis

Figure 9 and Table 3 stratify frozen results into predefined regimes: 2008–2009 crisis, 2010–2012 elevated volatility, 2013–2014 calm, 2015–2016 stress, 2017 calm, 2018 spike. This is an explanatory analysis of frozen results, not a basis for re-optimization.

16 Robustness

Seed robustness for the neural models (predeclared seeds 7, 42, 2026; headline series is the primary seed 42, not an ensemble): the canonical run tables (`../outputs/runs/e34928235ed6-058d5aea/tables/seed_robustness_summary.csv`). DM and block-bootstrap pairwise comparisons: Table 4.

Table 1: Frozen out-of-sample backtesting results by model and tail.

Tail 1%							
Model	**N**	**Viol**	**Rate**	**Kupiec p**	**Ind p**	**CC p**	**Pinball**
HS	2641	41	0.0155	0.0083	0.0000	0.0000	0.0006
GARCH-t	2641	40	0.0151	0.0135	0.1484	0.0167	0.0004
LinQR	2641	57	0.0216	0.0000	0.0411	0.0000	0.0005
MLP	2641	61	0.0231	0.0000	0.0654	0.0000	0.0004
GJR-t	2641	38	0.0144	0.0334	0.5766	0.0891	0.0003
GRU	2641	96	0.0363	0.0000	0.0002	0.0000	0.0007
Tail 5%							
Model	**N**	**Viol**	**Rate**	**Kupiec p**	**Ind p**	**CC p**	**Pinball**
HS	2641	140	0.0530	0.4819	0.0000	0.0002	0.0017
GARCH-t	2641	166	0.0629	0.0035	0.2304	0.0069	0.0013
LinQR	2641	161	0.0610	0.0123	0.0020	0.0004	0.0013
MLP	2641	152	0.0576	0.0816	0.0182	0.0135	0.0013
GJR-t	2641	167	0.0632	0.0027	0.3814	0.0075	0.0013
GRU	2641	178	0.0674	0.0001	0.0106	0.0000	0.0015
Tail 10%							
Model	**N**	**Viol**	**Rate**	**Kupiec p**	**Ind p**	**CC p**	**Pinball**
HS	2641	255	0.0966	0.5530	0.0006	0.0022	0.0024
GARCH-t	2641	300	0.1136	0.0224	0.2361	0.0365	0.0021
LinQR	2641	297	0.1125	0.0360	0.1042	0.0297	0.0021
MLP	2641	256	0.0969	0.5976	0.0016	0.0060	0.0022
GJR-t	2641	285	0.1079	0.1802	0.0648	0.0740	0.0020
GRU	2641	357	0.1352	0.0000	0.0395	0.0000	0.0022

17 Limitations

- The 1% tail has few expected violations (≈2.6 per year); coverage tests have low power.
- Neural models are retrained daily on at most 1500 observations; capacity is deliberately small.
- rv5/bv are daily aggregates; intraday dynamics are not modeled.
- DM/block-bootstrap inference at extreme tails relies on asymptotic approximations; results are interpreted as indicative.

18 Potential Improvements

Potential extensions include refitting the neural models on the full rolling window after selecting the stopping epoch, evaluating external validity on additional liquid assets, and considering more specialized dynamic quantile models (e.g., CAViaR-type specifications). These are deliberately left for future work: introducing them after inspecting the frozen test would violate the model-selection protocol followed in this study.

19 Conclusion

The conclusion is drawn from the frozen out-of-sample evidence in Section 12 and the analysis sections.

Table 2: Feature ablation: failure rate and mean pinball by information set.

Model	Tail	Rate	Pinball
M2-F0	1%	0.0170	0.0004
M2-F0	5%	0.0541	0.0014
M2-F0	10%	0.1037	0.0022
M2-F1	1%	0.0174	0.0004
M2-F1	5%	0.0557	0.0013
M2-F1	10%	0.1079	0.0021
M2-F2	1%	0.0216	0.0005
M2-F2	5%	0.0606	0.0013
M2-F2	10%	0.1121	0.0021
M2-F3	1%	0.0216	0.0005
M2-F3	5%	0.0610	0.0013
M2-F3	10%	0.1125	0.0021
M3-F0	1%	0.0265	0.0005
M3-F0	5%	0.0545	0.0014
M3-F0	10%	0.1026	0.0022
M3-F1	1%	0.0314	0.0005
M3-F1	5%	0.0545	0.0013
M3-F1	10%	0.0966	0.0022
M3-F2	1%	0.0273	0.0004
M3-F2	5%	0.0579	0.0013
M3-F2	10%	0.0988	0.0022
M3-F3	1%	0.0231	0.0004
M3-F3	5%	0.0576	0.0013
M3-F3	10%	0.0969	0.0022

At the 1% tail, the model with the empirical failure rate closest to the nominal level is GJR-t (0.0144 vs target 0.01); the lowest mean pinball loss is achieved by GJR-t (0.00034). Closeness of the unconditional failure frequency is not, by itself, evidence of adequate conditional VaR dynamics (see the coverage tests and the violation-clustering diagnostics below).

At the 5% tail, the model with the empirical failure rate closest to the nominal level is HS (0.0530 vs target 0.05); the lowest mean pinball loss is achieved by GJR-t (0.00126). Closeness of the unconditional failure frequency is not, by itself, evidence of adequate conditional VaR dynamics (see the coverage tests and the violation-clustering diagnostics below).

At the 10% tail, the model with the empirical failure rate closest to the nominal level is MLP (0.0969 vs target 0.10); the lowest mean pinball loss is achieved by GJR-t (0.00204). Closeness of the unconditional failure frequency is not, by itself, evidence of adequate conditional VaR dynamics (see the coverage tests and the violation-clustering diagnostics below).

The GARCH family (GARCH-t, GJR-t) achieves the lowest mean pinball loss at all three tails. At the 1% tail, GJR-t has the closest failure rate among the primary models (0.0144 vs 0.0151 for GARCH-t; both lie above the 1% nominal, i.e. mildly under-conservative, and unconditional coverage rejects the nominal 1% rate at the 5% level for both: Kupiec p=0.0135 / 0.0334). Conditional coverage passes only for GJR-t at the 1% tail (CC p=0.0891), while GARCH-t does not (CC p=0.0167); at the 5% tail both fail conditional coverage (CC p=0.0069 / 0.0075). At the 10% tail, GARCH-t fails the conditional-coverage test at the 5% level (CC p=0.0365) whereas GJR-t

Table 3: Failure rate by regime (primary configuration per model).

Regime	GARCH-t	GJR-t	GRU	HS	LinQR	MLP
calm_2013_2014	0.0556	0.0556	0.0536	0.0020	0.0575	0.0456
calm_2017	0.0239	0.0199	0.0279	0.0120	0.0239	0.0199
crisis_2008_2009	0.0772	0.0792	0.1327	0.1564	0.0772	0.0713
elevated_2010_2012	0.0690	0.0650	0.0504	0.0345	0.0663	0.0544
spike_2018	0.0894	0.0894	0.1220	0.1057	0.0976	0.1138
stress_2015_2016	0.0595	0.0675	0.0476	0.0357	0.0496	0.0655

Table 4: Pairwise loss comparisons (negative DM favors model_a).

Pair	Tail	DM	DM p	Holm p	Boot p	Favors	Headline
M1 vs M3	1%	-2.0587	0.0395	0.0790	0.0900	a	H
M1 vs M4	1%	2.2649	0.0235	0.0705	0.0420	b	H
M2 vs M3	1%	1.3397	0.1803	0.1803	0.4720	b	H
M3 vs M5	1%	-2.8785	0.0040	0.0160	0.2270	a	H
M1 vs M3	5%	-0.5033	0.6148	1.0000	0.6010	a	H
M1 vs M4	5%	3.1607	0.0016	0.0063	0.0010	b	H
M2 vs M3	5%	0.5166	0.6054	1.0000	0.6970	b	H
M3 vs M5	5%	-2.2169	0.0266	0.0799	0.3050	a	H
M1 vs M3	10%	-2.6016	0.0093	0.0227	0.0120	a	H
M1 vs M4	10%	2.7657	0.0057	0.0227	0.0030	b	H
M2 vs M3	10%	-2.7286	0.0064	0.0227	0.0250	a	H
M3 vs M5	10%	-0.6926	0.4886	0.4886	0.6990	a	H

does not (CC p=0.0740). At the 5% tail HS is marginally closer to the nominal frequency (0.0530 vs 0.0629). crisis-period 5% failure rates are 0.1564 (HS) vs 0.0772 (GARCH-t), documenting slow two-sided regime adaptation of historical simulation

The neural models do not deliver consistent absolute out-of-sample improvements over the classical baselines: the MLP versus GARCH-t is not significant at the 1% tail (Holm-corrected DM p=0.0790) and significant at the 10% tail (Holm p=0.0227), while the 5% difference is not statistically decisive (Holm p=1.0000); versus the linear quantile model it is significant at the 10% tail (Holm p=0.0227). The GRU is strongly under-conservative at the 1% tail (failure rate 0.0363, far above the 1% nominal), with the largest across-seed dispersion of the neural models. Under the pre-specified architecture family and development-only tuning, this is a valid negative result for this information set and sample; it should not be generalized to all neural VaR architectures.

Feature ablations isolate the incremental value of each realized-measure block by model and tail (relative pinball change; positive = improvement). Adding RV (F0 $\rightarrow$ F1): LinQR 1%: +0.1%, 5%: +4.4%, 10%: +4.8% - improvements at the 5%, 10% tails; MLP 1%: +8.2%, 5%: +2.0%, 10%: -0.6% - improvements at the 1%, 5% tails. Adding BV conditional on RV (F1 $\rightarrow$ F2): LinQR 1%: -1.3%, 5%: +0.1%, 10%: -0.5% - essentially no incremental gain to linear quantile regression once RV is observed; MLP 1%: +10.9%, 5%: +2.8%, 10%: +1.5% - the MLP appears able to exploit BV information, particularly at the 1% tail (+10.9%). This additional information value is nevertheless insufficient to make the MLP outperform GARCH/GJR in absolute forecast loss (information increment $\neq$ model-level superiority). The F3 jump/downside block (F2 $\rightarrow$ F3): LinQR 1%: -14.7%, 5%: -1.0%, 10%: -0.4%; MLP 1%: -3.9%, 5%: -0.7%, 10%: -0.1% - no stable positive contribution at any model-tail cell; the F3 effect is attributed to the block as a whole, not to any single component.

20 References

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A Full Pairwise DM Matrix

The complete 45-row pairwise Diebold–Mariano comparison matrix (exploratory; the body table shows only the predeclared headline pairs with Holm correction).

B Machine-Generated Data Audit

docs/DATA_AUDIT.md (generated by scripts/audit_data.py).

C Artifacts

All prediction panels: ../outputs/runs/e34928235ed6-058d5aea/predictions/*.parquet with sidecar manifests; metric tables: ../outputs/runs/e34928235ed6-058d5aea/tables/*.csv; figures: ../outputs/runs/e34928235ed6-058d5aea/figures/*.png. Freeze manifest: docs/FREEZE_MANIFEST.md.

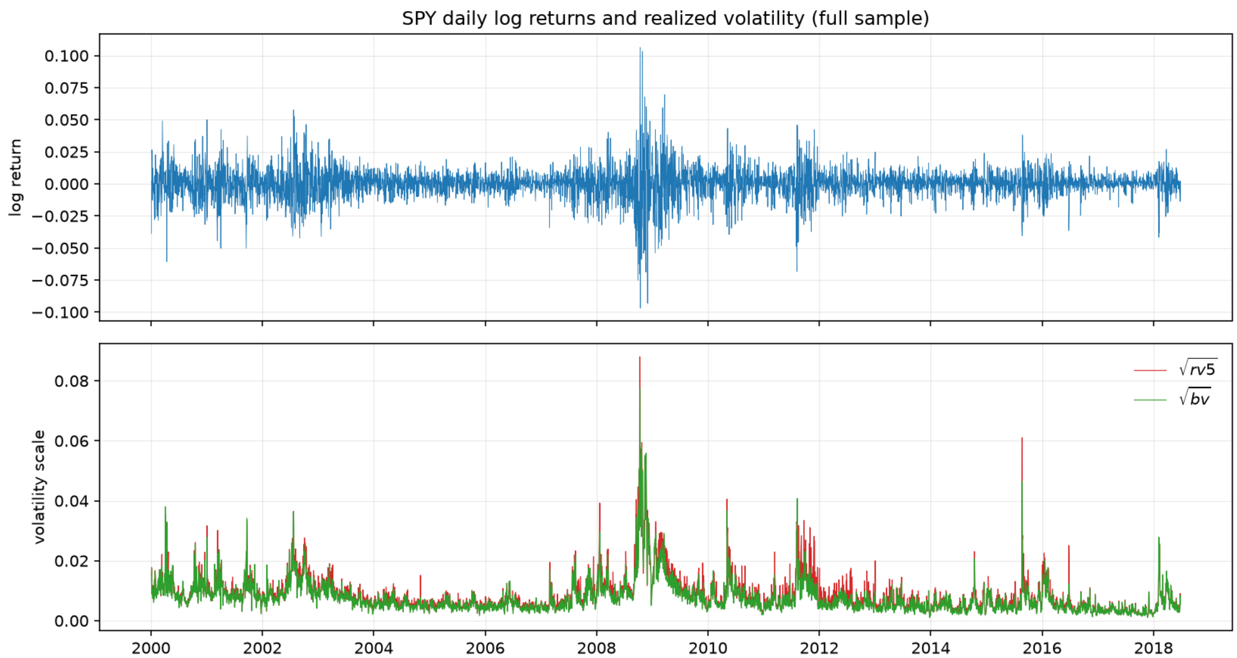

Figure 1: SPY log returns and volatility scale, full sample.

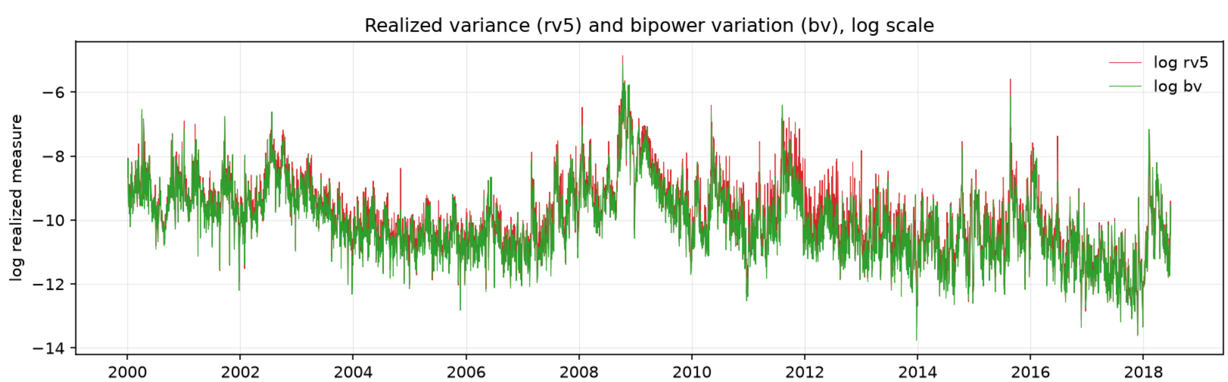

Figure 2: Realized-volatility measure and bipower variation (log scale).

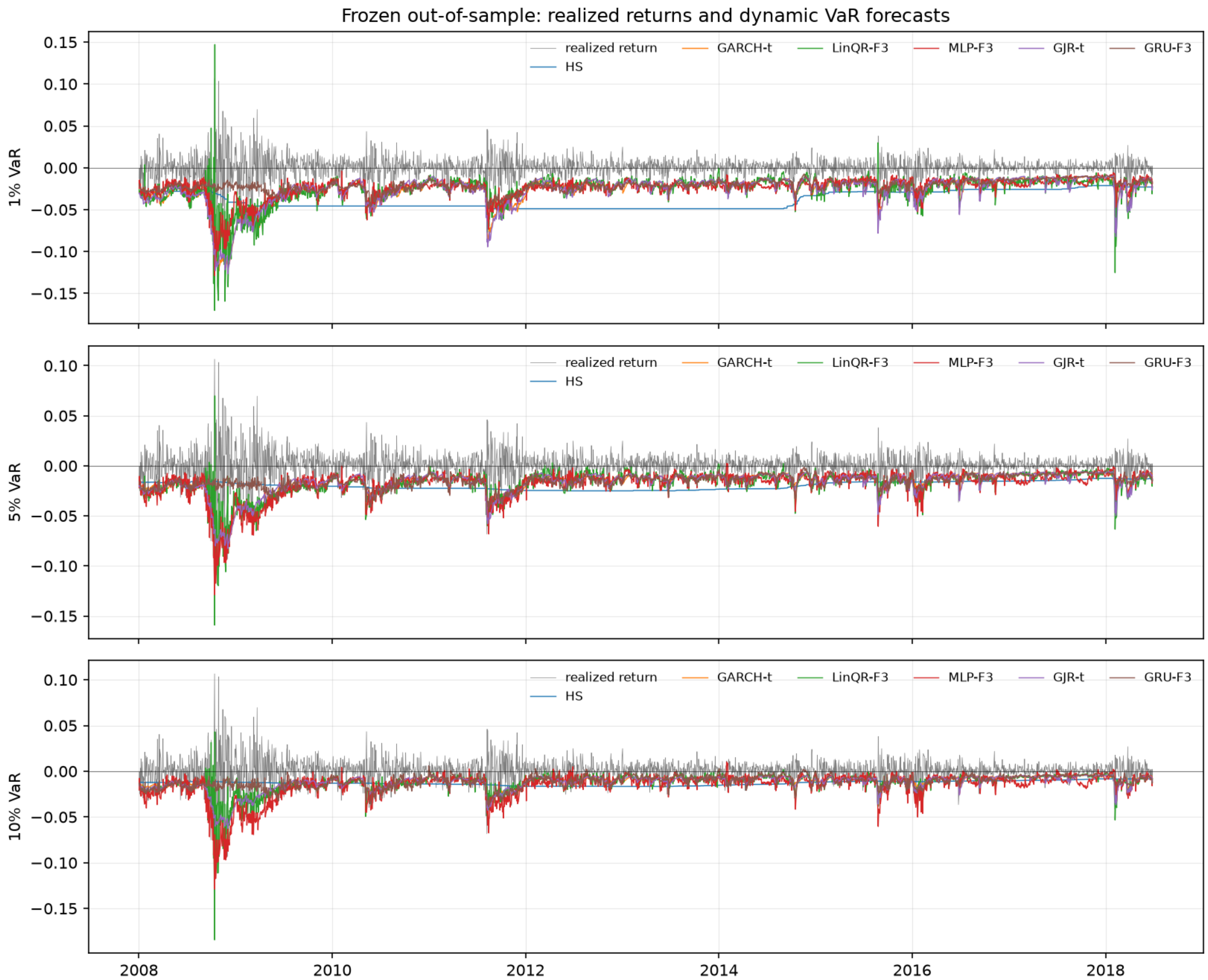

Figure 3: Frozen out-of-sample realized returns with dynamic VaR forecasts.

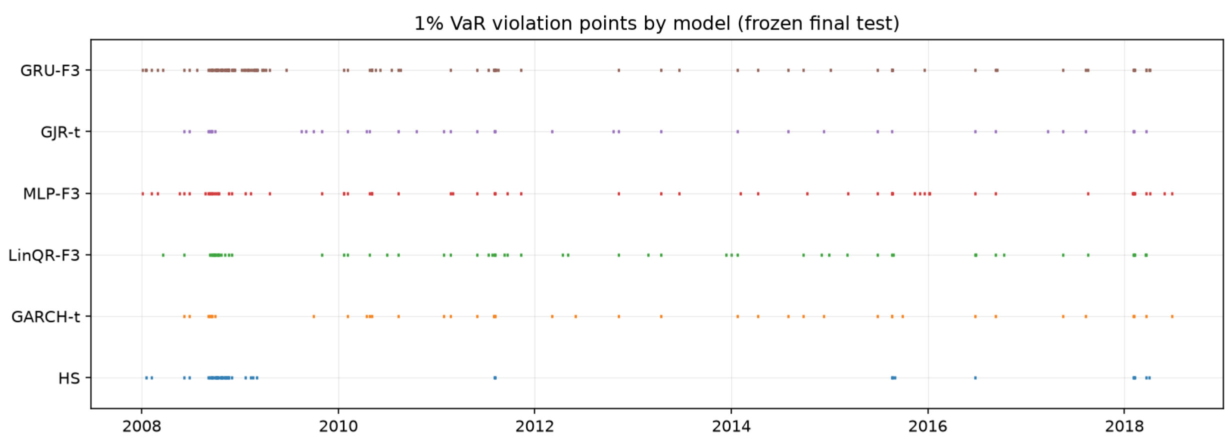

Figure 4: 1% VaR violation points by model.

Table 5: Full pairwise DM matrix (all model pairs, all tails). Headline pairs are marked H.

Pair	Tail	DM	DM p	Holm p	Boot p	Favors	Headline
M0 vs M1	1%	3.8448	0.0001	–	0.2020	b	
M0 vs M2	1%	2.0039	0.0451	–	0.1230	b	
M0 vs M3	1%	3.3399	0.0008	–	0.2090	b	
M0 vs M4	1%	4.0951	0.0000	–	0.1870	b	
M0 vs M5	1%	-1.3604	0.1737	–	0.4790	a	
M1 vs M2	1%	-1.9828	0.0474	–	0.3570	a	
M1 vs M3	1%	-2.0587	0.0395	0.0790	0.0900	a	H
M1 vs M4	1%	2.2649	0.0235	0.0705	0.0420	b	H
M1 vs M5	1%	-3.2243	0.0013	–	0.2080	a	
M2 vs M3	1%	1.3397	0.1803	0.1803	0.4720	b	H
M2 vs M4	1%	2.2050	0.0275	–	0.3340	b	
M2 vs M5	1%	-2.2740	0.0230	–	0.2170	a	
M3 vs M4	1%	2.4669	0.0136	–	0.0580	b	
M3 vs M5	1%	-2.8785	0.0040	0.0160	0.2270	a	H
M4 vs M5	1%	-3.3619	0.0008	–	0.1910	a	
M0 vs M1	5%	4.5636	0.0000	–	0.0770	b	
M0 vs M2	5%	4.7905	0.0000	–	0.0400	b	
M0 vs M3	5%	4.0315	0.0001	–	0.0910	b	
M0 vs M4	5%	4.8280	0.0000	–	0.0520	b	
M0 vs M5	5%	7.2079	0.0000	–	0.0010	b	
M1 vs M2	5%	-1.0079	0.3135	–	0.4600	a	
M1 vs M3	5%	-0.5033	0.6148	1.0000	0.6010	a	H
M1 vs M4	5%	3.1607	0.0016	0.0063	0.0010	b	H
M1 vs M5	5%	-2.6090	0.0091	–	0.2620	a	
M2 vs M3	5%	0.5166	0.6054	1.0000	0.6970	b	H
M2 vs M4	5%	1.9230	0.0545	–	0.2270	b	
M2 vs M5	5%	-2.5373	0.0112	–	0.2290	a	
M3 vs M4	5%	1.5973	0.1102	–	0.0980	b	
M3 vs M5	5%	-2.2169	0.0266	0.0799	0.3050	a	H
M4 vs M5	5%	-2.9635	0.0030	–	0.2140	a	
M0 vs M1	10%	4.8013	0.0000	–	0.0510	b	
M0 vs M2	10%	4.9154	0.0000	–	0.0300	b	
M0 vs M3	10%	2.7171	0.0066	–	0.1410	b	
M0 vs M4	10%	5.0337	0.0000	–	0.0290	b	
M0 vs M5	10%	5.3083	0.0000	–	0.0010	b	
M1 vs M2	10%	-0.1940	0.8462	–	0.8530	a	
M1 vs M3	10%	-2.6016	0.0093	0.0227	0.0120	a	H
M1 vs M4	10%	2.7657	0.0057	0.0227	0.0030	b	H
M1 vs M5	10%	-2.5975	0.0094	–	0.2490	a	
M2 vs M3	10%	-2.7286	0.0064	0.0227	0.0250	a	H
M2 vs M4	10%	1.0913	0.2751	–	0.3070	b	
M2 vs M5	10%	-2.7651	0.0057	–	0.1820	a	
M3 vs M4	10%	3.4390	0.0006	–	0.0020	b	
M3 vs M5	10%	-0.6926	0.4886	0.4886	0.6990	a	H
M4 vs M5	10%	-2.9867	0.0028	–	0.1840	a	

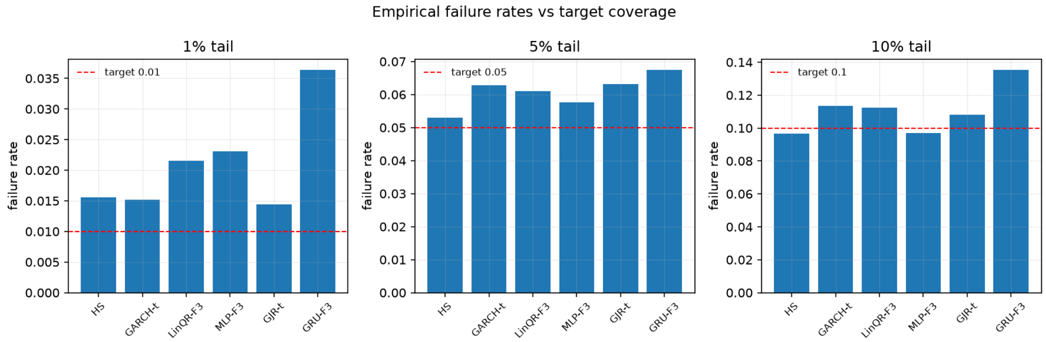

Figure 5: Empirical failure rates vs target coverage.

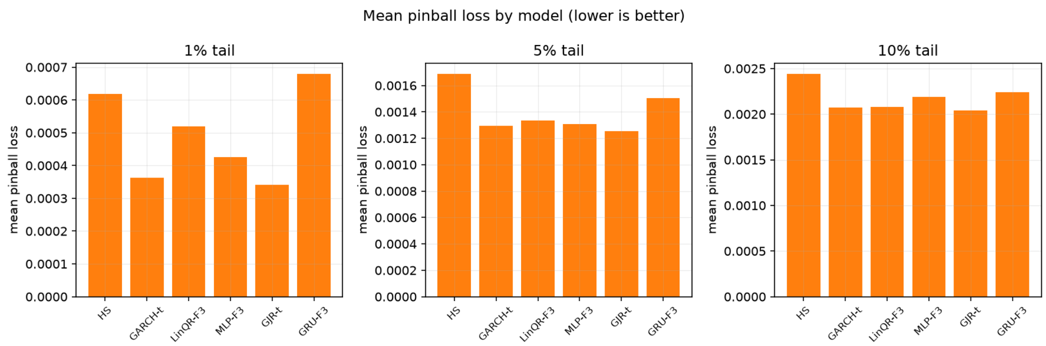

Figure 6: Mean pinball loss by model and tail.

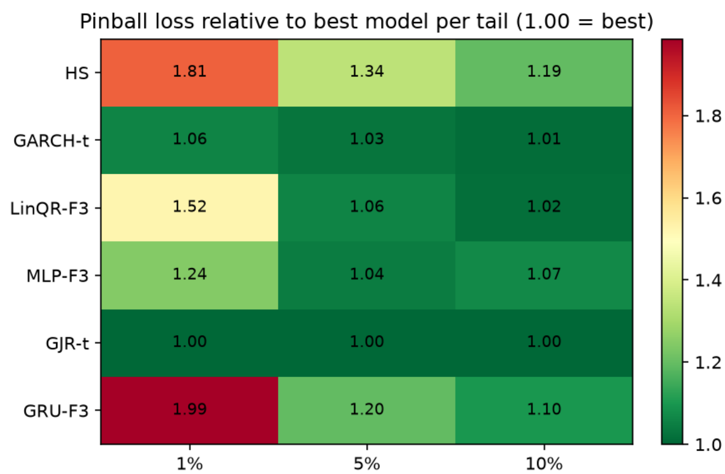

Figure 7: Pinball loss relative to best model per tail.

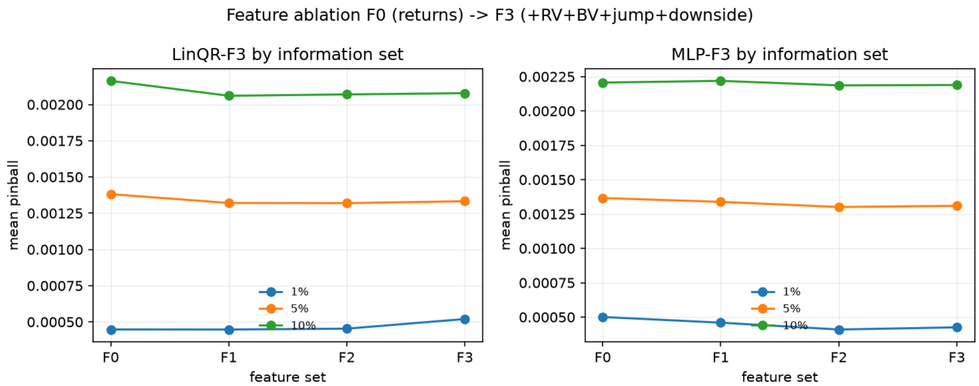

Figure 8: Feature ablation F0–F3 for linear quantile regression and MLP.

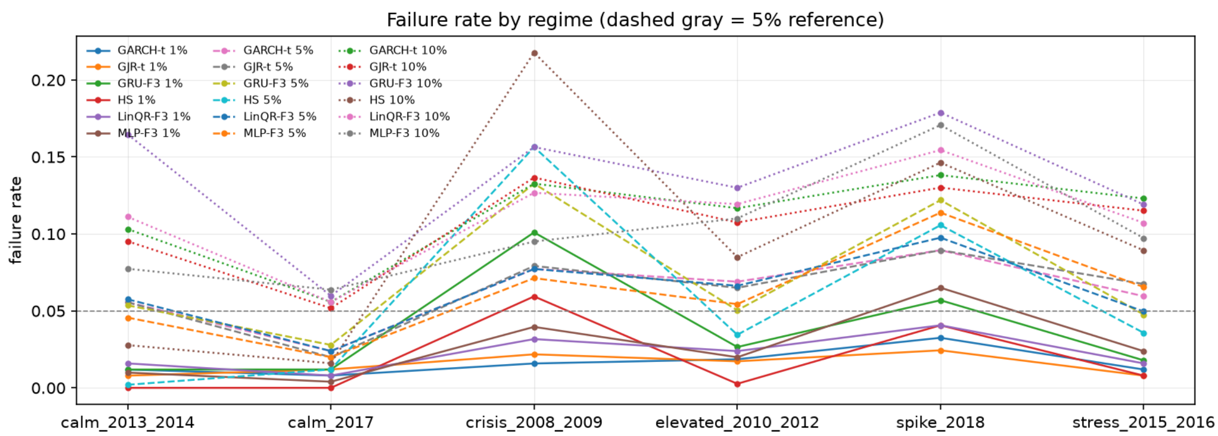

Figure 9: Failure rate by regime.

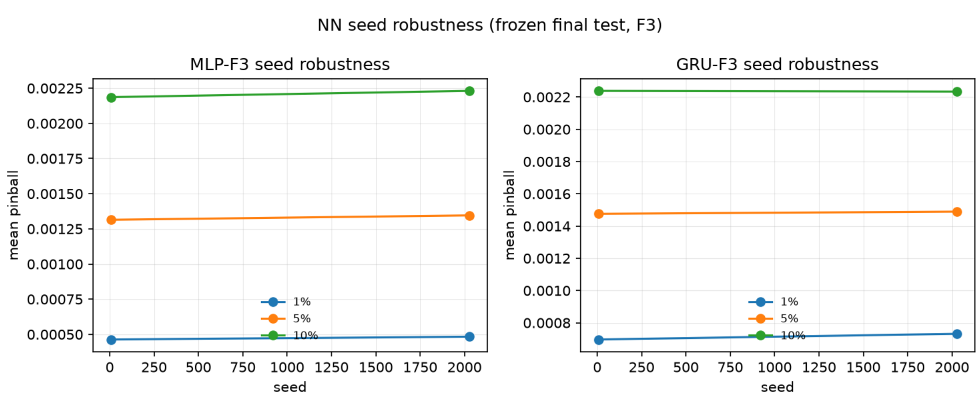

Figure 10: Neural seed robustness.